

# Thoughts on Some Connections Between General Relativity and the Knowledge of Observers

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## Abstract

In this note, the authors examine the relation between coordinate systems “covering” the universe and the observer who uses them.

The reference [1] is very captivating from the metaphysical angle. In this paper we take up reading [1] selectively and proceed along to some insights. The purpose is to broaden our outlook; general relativity is a horizon-broadening study. Figure 1.2 in [1] is interesting as it describes spacetime as being filled densely with worldlines which serve to locate events. The firing of an action potential is an event in spacetime. It is concomitant with ambient electric field changes that can influence the firing of action potentials at other spatiotemporal positions, i.e., set off other events potentially<sup>1</sup>. The action potential event is the reality, the coordinate system telling us where and when it occurred is arbitrary. Multiple coordinate systems can refer to the same action potential firing event and one can transform between these systems. We assume henceforth that there is no preferred coordinate system to follow, for measuring and recording the action potential firing event.

“In christening events with coordinates one demands smoothness but forgoes every thought of mensuration.”

Consider the event  $a$  that the action potential is initiated on axon 1 and the event  $b$  that it has been initiated on axon 2. Let  $(a1, a2, a3, a4)$  be the coordinates of event  $a$ . Let  $(b1, b2, b3, b4)$  be the coordinates of event  $b$ .

“The four numbers of an event are nothing but an elaborate telephone number.”

We can learn whether two events are neighbors from these two telephone numbers but learn nothing about their separation.

“Nothing prevents an event from being catalogued by alternative coordinate systems.”

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<sup>1</sup>These electric fields were studied in [2].

Suppose  $a$  and  $b$  are considered to be neighbours by the nearness of their coordinate values in a smooth coordinate system. Draw a little vector from  $a$  to  $b$ . The ‘name’ of this vector will be the quadruple of component differences between  $a$  and  $b$ . Having thus set the stage, consider:

“Two is the minimum number of coordinate patches [or coordinate systems] required to cover the two-sphere without singularity.”

This statement has implications. Does it mean that a single observer cannot see the whole universe? This might be valid, for he can only see outwards. If he could, in addition, also see himself, then the singularity would be avoidable. Alternately, another one viewing him should have a coordinate system and then the two would cover the whole universe, together. Both would be partially blind, but jointly omniscient. It reminds one of the parable of the seven blind men measuring an elephant. However, if one takes the position of monism, there cannot be a second observer. Hence there would always be a singularity in the single coordinate system used to observe the universe. Which means meditation on the universe would always leave you ignorant of some aspect of it. But this is true if there is a duality between the meditator and the universe. If however, the two become fused, that is, one object<sup>2</sup>, then there is full knowledge alone – no holder of that knowledge anymore. To summarize, you can have information that there is a knower and there is a known, but then you don’t have knowledge. Alternately, you can have knowledge, but then you are dissolved.

## References

- [1] Kip S Thorne, Charles W Misner, and John Archibald Wheeler. *Gravitation*. Freeman, 2000.
- [2] Aman Chawla, Salvatore D. Morgera, and Arthur D. Snider. Fields, geometry, and their impact on axon interaction. *Journal of Applied Mathematics and Physics*, 9(4):751–778, 2021.

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<sup>2</sup>Or, more appropriately, one subject.